

EXPERIMENT No. - 6

Date:

**TO IDENTIFY THE FUNCTIONAL GROUP PRESENT IN THE GIVEN ORGANIC COMPOUND
(Sample F₁)****APPARATUS REQUIRED**

- | | |
|---------------------------|--------------------------|
| a) Test tubes..... | b) Beakers..... |
| c) Bunsen..... | d) Test tube holder..... |
| e) Glass rod / pipes..... | f) Watch glass..... |
| g) Filter paper..... | h) Litmus paper..... |
| i) Spatula..... | j) |

CHEMICALS REQUIRED

- | | |
|----------------------------------|--|
| a) Given organic compound..... | b) Baeyer's reagent..... |
| c) NaHCO_3 | d) FeCl_3 |
| e) 2,4-DNP..... | f) $(\text{NH}_4)_2\text{Ce}(\text{NO}_3)_6$ |
| g) H_2SO_4 | h) NaOH |
| i) | j) |

THEORY

Hydrocarbons and their derivatives in which covalently bonded carbon is the essential element are called organic compounds. E.g: $\text{C}_2\text{H}_5\text{OH}$, CH_3COOH , C_6H_6 , CH_4 , NH_2CONH_2 etc.

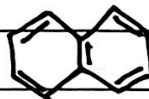
Cyclic compounds having one or more benzene rings in their molecules are called aromatic hydrocarbons. E.g:



Benzene



phenol



naphthalene

Open-chain compounds having no rings of any type are called aliphatic compounds. E.g.: C_2H_6 , C_2H_5OH , CH_4 etc.

Organic compounds having no carbon-carbon double or triple bonds are called saturated compounds. E.g.: CH_3-CH_3 , $CH_3-CH_2-CH_3$ etc.

Organic compounds having at least one carbon-carbon double or triple bond are called unsaturated compounds. E.g.: $CH_2=CH_2$, $CH\equiv CH$ etc.

An atom or a group of atoms attached to carbon atoms in an organic compound that determines the chemical properties of that compound is called a Functional group.

E.g.: $-OH$ (alcohol), $-NO_2$ (nitro), $-CHO$ (aldehyde), $-COOH$ (carboxylic acid), etc.

A functional group characterized by a carbon atom double-bonded to an oxygen atom ($C=O$) is carbonyl group.

A quick, general, non-specific test used to screen a sample for the likely presence of carbonyl group without identifying the exact type of molecule is preliminary test.

Test used to verify the specific type of carbonyl compound present is confirmation test.

1. PRELIMINARY EXAMINATIONS:

- a. Physical State: Solid
- b. Colour: Colourless
- c. Odour or Smell: characteristic odour

2. Solubility Test (+ve for soluble and -ve for insoluble)

Cold water	Hot water	Dil. NaOH	Dil. HCl	Dil. NaHCO ₃	Organic Solvent
+ve	+ve	+ve	+ve	+ve	+ve

3. Litmus Test:

S.N.	Experiments	Observations	Inferences
1.	Both red and blue litmus papers were dipped in aqueous solution of the organic sample.	Blue litmus was turned to red.	May be carboxylic acid or phenol.

4. Ignition Test

S.N.	Experiments	Observations	Inferences
1.	A pinch of organic sample was taken in a spatula and was burnt over the flame of burner.	Non-sooty or non-smoky flames were seen.	May be aliphatic compound with low carbon percentage

5. Unsaturation Test

S.N.	Experiments	Observations	Inferences
1.	Baeyer's Test (Alkaline KMnO_4) Few drops of organic sample were taken in a test tube and few drops of very dilute alkaline KMnO_4 solution were added.	Pink colour of KMnO_4 was not discharged.	May be saturated compound.

6. Preliminary test for functional group:

S.N.	Experiments	Observations	Inferences
1.	Sodium bicarbonate (NaHCO_3) Test: 2 ml of cold dilute NaHCO_3 solution was taken in a test tube and few drops of organic sample were added.	Brisk effervescence of CO_2 was seen.	May be presence of Carboxylic group.

2.	Ferric chloride (FeCl_3) Test: 1-2 ml of FeCl_3 solution was taken in a test tube and few drops of organic sample were added.	Green or violet or blue colouration was not seen.	May be absence of phenol group.
3.	2,4 - DNP Test 1-2 ml of 2,4 - DNP solution was taken in a test tube and few drops of an organic sample were added.	yellow or orange-red ppt was not seen.	May be absence of carbonyl group.
4.	Cerric Ammonium Nitrate test: 1 ml of $(\text{NH}_4)_2\text{Ce}(\text{NO}_3)_6$ solution was taken in a test tube and Few drops of organic sample were added.	Wine red or pink colouration was not seen.	May be absence of alcohol group.

5.	Carbylamine Test 1 ml of alc. KOH along with 3-4 drops of Chloroform was taken in a test tube and a few drops of organic sample were added.	Highly offensive odour was not observed	May be absence of primary amines.
6.	Dilute HCl Test 1 ml of dil. HCl was taken in a test tube and few drops of organic sample were added.	The substance was not dissolved.	May be absence of amines.

TEST REACTION

Sodium bicarbonate test

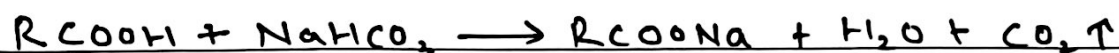


7. Confirmatory Tests for ...Carboxylic group...(-COOH).....

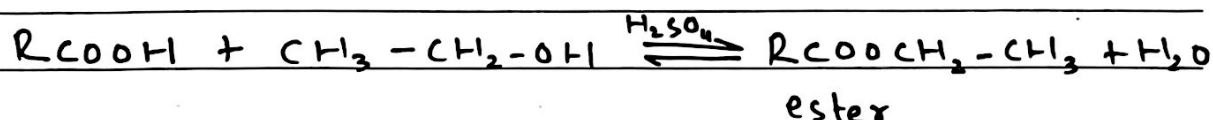
S.N.	Experiments	Observations	Inferences
①	Sodium bicarbonate test:		
	- 1 ml of aq. soln of the sample was taken in a test tube & cold saturated soln of sodium bicarbonate was added drop by drop.	Vigorous eff- erescence of CO ₂ gas was seen.	Presence of carboxylic group (-COOH)
②	Esterification Test:		
	- 1 ml soln of the sample was taken in a test tube & 1 ml of ethanol and few drops of conc. H ₂ SO ₄ were added, and the tube was heated on hot water bath.	pleasant fruity fruity smell was observed.	Presence of carboxylic group (-COOH)
③	Sodium hydroxide test:		
	- The organic sample was dissolved in dil. NaOH and dil. HCl was added into it.	Compound was dissolved and reappeared on the addition of HCl.	Presence of carboxylic group (-COOH)

TEST REACTIONS

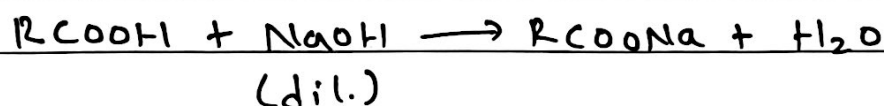
1. Sodium bicarbonate test:



2. Esterification test:



3. Sodium hydroxide test:



RESULT

In this way, the Functional group present in the given organic compound was found to be carboxylic acid group ($-\text{COOH}$).

CONCLUSION

Hence, From the above experiment, it can be concluded that the functional group in the given organic compound can be detected by preliminary and distinctive tests.

PRECAUTIONS

1. Personal protective equipment should be worn.
2. Organic compounds with reagent should not be heated directly on Flame but water bath should be used.
3. Organic compound should be used in minimum quantity.
4. Apparatus should be handled with care.

VIVA-VOCE

1. What is functional group?
2. What is Tollen's reagent?
3. Does phenol respond to the sodium bicarbonate test like carboxyl groups?
4. Why is esterification reaction carried out in the presence of concentrated sulphuric acid?
5. How do phenols differ from alcohols although both contain $-OH$ group?
6. What is Baeyer's reagent?
7. Name a compound which gives tests for both the carboxylic as well as aldehydic group.

1. An atom or a group of atoms attached to carbon atoms in an organic compound that determines its chemical properties is called a Functional group. Eg: $-COOH$, $-OH$

2. Tollen's reagent is ammoniacal silver nitrate solution $[Ag(NH_3)_2OH]$.

3. Phenol being weaker than carboxylic acid does not respond to the test.

4. Conc. H_2SO_4 acts as catalyst as well as the dehydrating agent.

5. Phenols are acidic and turn blue litmus to red. Alcohols are neutral & have no effect on litmus.

6. Baeyer's reagent is 1% cold solution of potassium permanganate in water, made alkaline with sodium carbonate

7. Formic / methanoic acid ($HCOOH$).